

TASK RECORD

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TASK: OSI MODEL, TCP AND UDP

Task Description:

The objective of this task is to study and understand the basic concepts of computer networking, especially the OSI model, TCP, and UDP.

INTRODUCTION:

A computer network allows different devices to connect, communicate and share data. There are different models and protocols that help us understand how data travels through a network. The OSI model explains network communication in seven layers. TCP and UDP are two protocols used for sending and receiving data between devices.

OSI MODEL:

OSI means Open Systems Interconnection. It is a model used to understand how communication happens between computers in a network. The OSI model has seven layers. Each layer has its own work and helps in sending data from one device to another.

The seven layers are:

1. Application Layer
2. Presentation Layer
3. Session Layer
4. Transport Layer
5. Network Layer
6. Data Link Layer
7. Physical Layer

STRUCTURE:

Physical layer



Data link layer



Network layer



Transport layer



Session layer



Presentation layer



Application layer

1. Application Layer:

This is the top layer of the OSI model. It helps the user to use network services.

For example, when we use a browser, email, or file transfer, this layer is used.

Examples: SMTP(simple mail transfer protocol), DNS(domain name system).

2. Presentation Layer:

This layer takes care of the way data is shown or understood. It can also change the data format, reduce the size of data, and protect the data using encryption.

work:

1. Changes data format
2. Encrypts data and Compresses data.

3. Session Layer:

This layer helps to start and control communication between two devices. It keeps the communication active and closes it when the communication is finished.

work:

1. Starts communication
2. Maintains communication
3. Ends communication

4. Transport Layer:

This layer helps to send data from one device to another. It also checks whether the data reaches properly.

The two main protocols used here are TCP and UDP.

work:

1. Sends data
2. Checks errors
3. Divides data into smaller parts

5. Network Layer:

This layer finds a path for the data to travel from the sender to the receiver. It mainly works with IP addresses.

work:

1. Finds the path
2. Uses IP address

Example: IP

6. Data Link Layer:

This layer helps to send data between devices that are directly connected. It also checks for errors in the data.

work:

1. Checks errors
2. Uses MAC addresses

7. Physical Layer:

This is the bottom layer of the OSI model. It deals with sending data as signals through cables or wireless connections.

Examples:

1. Network cable
2. Wireless signals

TCP:

TCP stands for Transmission Control Protocol. It is used to send data between computers in a safe and reliable way. TCP first creates a connection between the sender and receiver. Then it sends the data. It also checks whether the data is received or not.

Uses of TCP:

TCP is commonly used for:

1. Web browsing
2. Email
3. File transfer
4. Online banking

UDP:

UDP stands for User Datagram Protocol. It is used to send data quickly from one device to another. Unlike TCP, UDP does not create a connection before sending data. It also does not check whether every packet is received.

Uses of UDP:

UDP is commonly used for:

1. Online games
2. Video streaming
3. Voice calls
4. Live broadcasts
5. DNS